

Numerical differentiation

Approximation Formulae

⇒ Forward-Endpoint $h > 0$

$$f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h}, \text{ where } h = x_1 - x_0 =$$

$$\Rightarrow \text{Error Bound} = \frac{h}{2} f''(\xi)$$

⇒ Backward-Endpoint $h < 0$

$$f'(x_0) = \frac{f(x_0+h) - f(x_0)}{-h}, \text{ where } h \text{ should be -ve}$$

$$\Rightarrow \text{Error Bound} = \frac{h}{2} f''(\xi)$$

⇒ Three point formulae $\times$ Can use forward or backward

$$\text{Three point - Endpoint} \Rightarrow f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$$

$$\text{Error. B} = \frac{h^2}{3} f'''(\xi)$$

$$\text{Three point - Midpoint} \Rightarrow f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$\text{Error. B} = \frac{h^2}{6} f'''(\xi), \text{ using } x_0-h \text{ and } x_0+h$$

⇒ Five point formulae $\times$ Can use forward or backward

$$\text{Five point - Endpoint} \Rightarrow f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$\text{Error. B} = \frac{h^4}{5} f^{(5)}(\xi)$$

$$\text{Five point - Midpoint} \Rightarrow f'(x_0) = \frac{1}{12h} [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$\text{Error. B} = \frac{h^4}{30} f^{(5)}(\xi)$$

Q1%

1. Use the forward-difference formulas and backward-difference formulas to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
0.5	0.4794	0.852 $\rightarrow f$
0.6	0.5646	0.852 $\rightarrow B$
0.7	0.6442	0.796 $\rightarrow B$

$h=0.1$

forward % $h=0.1$

$$f'(0.5) = \frac{f(0.6) - f(0.5)}{0.1} = 0.852$$

$$f'(0.6) = \frac{f(0.7) - f(0.6)}{0.1} = 0.796$$

$$f'(0.7) = \frac{f(0.8) - f(0.7)}{0.1} = \text{Can't be solved bc } 0.8 \text{ not in table.}$$

Backward % $h=-0.1$

$$f'(0.5) = \frac{f(0.5+(-0.1)) - f(0.5)}{-0.1} = \frac{f(0.4) - f(0.5)}{-0.1} = \text{Can't be solved bc } 0.4 \text{ not in table.}$$

$$f'(0.6) = \frac{f(0.5) - f(0.6)}{-0.1} = 0.852$$

$$f'(0.7) = \frac{f(0.6) - f(0.7)}{-0.1} = 0.796$$

Q3%

3. The data in Exercise 1 were taken from the following functions. Compute the actual errors in Exercise 1, and find error bounds using the error formulas.

a. $f(x) = \sin x$

① Find $f'(x)$ and $f''(x)$ %

$$f'(x) = \cos x$$

$$f''(x) = -\sin x \quad (\text{'ll use for E.Bound})$$

② Actual Error % $\left| \text{Actual.E} - \text{Apper.E} \right|$ from Q1

$$\textcircled{1} \Rightarrow f(0.5) = \cos(0.5) = 0.8775$$

$$f'(0.6) = 0.8253$$

$$f'(0.7) = 0.7648$$

$$\textcircled{2} \text{ A.E for } (0.5) = |0.852 - 0.8775| = \underline{0.0255}$$

$$(0.6) = |0.852 - 0.8253| = \underline{0.0267}$$

$$(0.7) = |0.796 - 0.7648| = \underline{0.0312}$$

$$\textcircled{3} \text{ find E. Bound 8 } \left| \frac{h}{2} f''(\xi) \right|$$

$$\text{for } 0.5 \text{ } [0.5, 0.6] = \left| \frac{0.1}{2} (-\sin(\xi)) \right| = |0.05 (-\sin(\xi))| = |-0.05 \times 0.5646| = \underline{0.02823}$$

$$\text{for } 0.6 \text{ } [0.6, 0.7] = |-0.05 \times 0.6442| = \underline{0.03221}$$

$0.5 \leq \sin(\xi) \leq 0.6$ ← Substituted 0.6 and 0.5 in sin
 $\downarrow$
 0.4794 0.5646

$$\text{for } 0.7 \text{ } [0.6, 0.7] = \underline{0.03221}$$

$0.6 \leq \sin(\xi) \leq 0.7$
 $\downarrow$ 0.6442

Q58

5. Use the most accurate three-point formula to determine each missing entry in the following tables.

d.

	x	$f(x)$	$f'(x)$
Forward ←	x_0 2.0	3.6887983	
mid-Point ←	x_1 2.1	3.6905701	
Backward ←	x_2 2.2	3.6688192	
	x_3 2.3	3.6245909	

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$$

$$\text{Error: } B = \frac{h^2}{3} f'''(\xi)$$

$$\text{Three-point - Midpoint: } f'(x_1) = \frac{1}{2h} [f(x_2+h) - f(x_0-h)]$$

, Error: $B = \frac{h^2}{6} f'''(\xi)$, using x_2+h and x_0-h

$$h = 0.1$$

Three-point - Endpoint :

Endpoint (Forward) : $h > 0$

$$\hat{f}'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)] = 0.1353$$

$$\hat{f}'(x_1) = -0.1051$$

Endpoint (Backward) : $h < 0 \Rightarrow -0.1$

$$\hat{f}'(2.2) = \frac{1}{2(-0.1)} [-3\overset{2.2}{f(x_2)} + 4\overset{2.1}{f(x_2+h)} - \overset{2.0}{f(x_2+2h)}]$$

$$= -0.3351$$

$$\hat{f}'(2.3) = -0.5546$$

Three-point - Midpoint : $\nearrow$ must use mid point only " x_1, x_2 "

$$\hat{f}'(x_1) = \frac{1}{2h} [f(\overset{2.2}{x_1+h}) - f(\overset{2.1}{x_1-h})] = -0.09989$$

$$\hat{f}'(x_2) = -0.329896$$